

LATENT PLANNING VIA EMBEDDING ARITHMETIC: A CONTRASTIVE APPROACH TO STRATEGIC REASONING

Anonymous authors
Anonymous submission

ABSTRACT

We study whether planning can be performed directly in an evaluation-aligned embedding space, without adding a policy head, value head, or learned dynamics model. We introduce SOLIS, a supervised-contrastive approach that maps chess positions so that empirical outcome similarity is represented by proximity and a single global *advantage vector* orders the space from losing to winning regions. At inference, SOLIS scores each candidate action by its alignment with that vector, turning search into a small number of latent-space comparisons. In chess experiments, this simple comparison rule produces favorable quantitative patterns: with 5-ply search under 50 ms/move conditions, SOLIS reaches 2500+ Elo, and in equal-depth matches at $S \in \{3, 4, 5\}$ it remains competitive with the configured Stockfish baseline while evaluating far fewer nodes. The results indicate that much of the practical benefit of search can be recovered from a well-structured representation, making evaluation-aligned latent planning a lightweight empirical alternative to dynamics modeling or policy learning.

1 INTRODUCTION

Planning in high-dimensional state spaces remains difficult because each additional decision step multiplies the number of candidate continuations that must be compared. Traditional reinforcement learning methods address this challenge through explicit search or learned policies, but both can become expensive as the action space expands (Jiang et al., 2023; Gieslmann and Pokorný, 2022). Latent planning offers a more compact comparison regime by evaluating decisions in learned representations rather than raw states, reducing the number and cost of the quantities that must be ranked (Hafner et al., 2019; 2020; Schrittwieser et al., 2020).

We ask whether the representation itself can supply the quantitative signal needed for planning. To test this, we introduce **SOLIS** (*Strategic Optimization via Latent Interpolative Spaces*), a chess engine whose transformer encoder is trained with supervised contrastive learning to place positions in an evaluation-aligned latent space. The resulting geometry makes two comparisons explicit: distance measures similarity in Stockfish-evaluated outcome, and projection onto a global “advantage vector” measures movement from losing regions toward winning regions. SOLIS uses this structure directly, ranking legal moves by the degree to which their child embeddings align with the advantage direction.

The empirical comparisons are favorable despite the deliberately small search budget. With only 5-ply search under 50 ms/move blitz conditions, SOLIS reaches an Elo of 2500+. In equal-depth matches at $S \in \{3, 4, 5\}$ with fixed branching factor $W=3$, SOLIS remains competitive with the configured Stockfish baseline, and the node counts show a practical margin in search effort: SOLIS evaluates 40–364 nodes while Stockfish averages hundreds of nodes and can reach thousands at the same depths. Game visualizations provide a complementary pattern, with decisive games tracing smooth paths toward the corresponding winning regions of the embedding.

This comparison structure distinguishes SOLIS from latent planners that pair learned dynamics with policy or value heads (Hafner et al., 2019; 2020; Schrittwieser et al., 2020). SOLIS instead asks how far action candidates move within an evaluation-aligned geometry, making the measured advantage direction the central decision statistic. Chess provides a controlled testbed with reliable state evaluations, but the same framing may apply to other perfect-information domains where comparable evaluation targets are available or can be learned.

2 BACKGROUND

Large action spaces make planning quantitatively demanding: each additional decision step expands the set of trajectories that must be scored, ranked, or pruned. We therefore frame the background around the comparison signals available to a planner—models, policies, value estimates, and representations—and then position SOLIS as a method that replaces broad rollout evaluation with low-cost latent comparisons.

2.1 MARKOV DECISION PROCESSES

Sequential decision-making tasks are commonly formalized as Markov Decision Processes (MDPs), represented by the tuple $\{\mathcal{S}, \mathcal{A}, T, R, p(s_0), \gamma\}$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, T is the transition model, R is the reward function, $p(s_0)$ is the initial state distribution, and $\gamma \in [0, 1]$ discounts future rewards (Puterman, 2014). A policy $\pi : \mathcal{S} \rightarrow p(\mathcal{A})$ then supplies the comparison distribution over actions available at each state.

Within an MDP, a *model* provides reversible access to the dynamics: for a given state–action pair, it returns the next state and reward (Moerland et al., 2022). Board games such as chess and Go supply this transition model exactly through their rules, whereas less specified environments require the dynamics to be estimated from data (Gieselmann and Pokorný, 2022).

Planning and reinforcement learning differ in how this dynamic information is used (Moerland et al., 2022). Planning methods simulate candidate futures from the current state and choose the highest-ranked local action, an approach analogous to mental lookahead (Russell and Norvig, 1995); the resulting decision is discarded after execution. Reinforcement learning methods typically do not assume reversible access to the dynamics (Sutton and Barto, 1998). They instead collect experience step by step and learn a *global solution*, such as a policy or value function, whose scores can be reused across the state space.

Model-based reinforcement learning combines these two sources of quantitative signal by using a model while also learning a reusable global solution. Following Moerland et al. (2022), the comparison is between two cases: learned-model settings such as Dyna (Sutton, 1991), where both dynamics and the global solution are estimated, and known-model settings such as AlphaZero (Silver et al., 2017), where the rules are given and learning focuses on the policy or value components.

2.1.1 MODEL-BASED RL WITH A LEARNED MODEL

A large body of prior work learns both the dynamics and the global decision statistic. PlaNet (Hafner et al., 2019) learns a latent dynamics model and pairs it with a learned global solution. Dreamer (Hafner et al., 2020), DreamerV2 (Hafner et al., 2022), and DreamerV3 (Hafner et al., 2024) scale this recipe to more complex tasks while retaining learned value functions and policies. MuZero (Schrittwieser et al., 2020) also learns latent dynamics, but adds tree search so that local comparisons can improve a learned policy and value function. Dyna (Sutton, 1991) and MBPO (Janer et al., 2021) provide related examples in which learned models improve value or policy learning.

2.1.2 MODEL-BASED RL WITH A KNOWN MODEL

When the environment dynamics are known, the empirical question shifts from modeling transitions to deciding how search estimates and learned scores should be combined. AlphaZero (Silver et al., 2017) uses exact game rules as the model and compares moves with Monte Carlo tree search (MCTS) guided by learned policy and value heads. SAVE (Hamrick et al., 2020) likewise assumes a simulator and links Q-learning with MCTS: the learned Q-function provides the search prior, and search-derived Q-estimates are amortized back into the network.

2.1.3 LATENT PLANNING

Latent planning reduces the dimensionality of these comparisons by operating in a compressed representation rather than the raw state space. PlaNet (Hafner et al., 2019), Dreamer (Hafner et al., 2020), and MuZero (Schrittwieser et al., 2020) combine latent dynamics with value or policy heads, so action choice still depends on learned evaluators after latent simulation. Expansive Latent Space

Trees (ELAST) (Gieselmann and Pokorny, 2022) use contrastive representations with sampling-based expansion for long-horizon control, while Trajectory Autoencoding Planner (TAP) (Jiang et al., 2023) learns a compact latent action space and plans over discrete action codes.

SOLIS removes these auxiliary evaluators from the comparison loop. It does not train a policy head, value head, or dynamics model; instead, it distills evaluation targets into the geometry of the representation, so inference compares moves by how strongly their embeddings move toward favorable regions.

2.2 CHESS ENGINES

Chess engines provide a useful quantitative baseline because strength, search depth, and node counts can be compared directly. IBM’s Deep Blue demonstrated the power of scaled chess search in 1997 by defeating reigning world champion Garry Kasparov. Early Stockfish versions (Romstad et al., 2008) followed the same broad pattern: handcrafted evaluation functions were paired with deep alpha-beta search. This produced superhuman play, but the quality of the static evaluation was limited by the human chess expertise encoded into it.

Those evaluation bottlenecks motivated systems that improved the scoring function while keeping comparisons among candidate moves central. AlphaZero (Silver et al., 2017) and Leela Chess Zero (Pascutto and Linscott, 2018) learned position evaluation through self-play reinforcement learning with no chess knowledge beyond the rules. Modern Stockfish versions adopted an efficiently updatable neural-network evaluation module (NNUE) (Nasu, 2018), improving evaluation speed and quality while retaining traditional alpha-beta search.

Although these engines far exceed human strength, their strongest results still rely on extensive search. The core principles trace back to v. Neumann (1928) and Shannon (1950), were later embodied in Deep Blue, and often require alpha-beta exploration over millions of states (Lai, 2015). Our experiments ask whether contrastive learning can shift some of this quantitative burden from breadth of search to quality of representation by planning as traversal through a high-dimensional embedding space.

2.3 CONTRASTIVE LEARNING

Contrastive learning provides the mechanism for turning evaluation labels into a geometry of comparisons. It trains a model to place similar inputs closer together and dissimilar inputs farther apart in latent space (Chen et al., 2020). Given a batch I with anchor input x_i , an encoder $f(\cdot)$ maps the anchor to a normalized embedding $z_i = f(x_i)$. Positive pairs (z_i, z_j) represent inputs that should be close under the target notion of similarity, while the remaining batch samples serve as negatives.

A standard way to quantify this relative similarity is the InfoNCE objective (van den Oord et al., 2019):

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \in A(i)} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (1)$$

Here, $A(i) = I \setminus \{i\}$ denotes the batch samples excluding anchor i , $\text{sim}(\cdot, \cdot)$ is the similarity metric, and τ controls the sharpness of the comparison. In self-supervised contrastive learning, the positive z_j is usually an augmentation of z_i , so the relative ranking is learned without labels (Chen et al., 2020; Khosla et al., 2021).

When labels are available, supervised contrastive learning (SupCon) (Khosla et al., 2021) strengthens the comparison by allowing each anchor to have multiple positives. The resulting loss is:

$$\mathcal{L}_{\text{sup}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a \in A(i)} \exp(\text{sim}(z_i, z_a)/\tau)}, \quad (2)$$

In this expression, $P(i)$ is the set of positives for anchor i , and $A(i)$ is the same comparison set as above. Prior work shows that increasing the number of positive and negative pairwise comparisons can improve representation quality (Khosla et al., 2021; Chowdhury et al., 2024).

3 METHODS

We next specify the components that determine the empirical comparisons in SOLIS: the tokenized chess state, the encoder that produces embeddings, the supervised-contrastive training signal, and the inference rule that converts latent geometry into move rankings.

3.1 INPUT REPRESENTATION AND TOKENIZATION

Each chess position is represented by its Forsyth-Edwards Notation (FEN) ASCII string, which records piece placement, side to move, castling rights, en passant target, half-move clock, and full-move counter in a compact format. Following Ruoss et al. (2024), we expand run-length encodings and tokenize each FEN into a fixed-length sequence of 77 tokens, giving every compared position the same input length. As noted by Ruoss et al. (2024), this representation omits some historical information that can affect special rules such as threefold repetition and the fifty-move rule, making the problem non-Markovian; we return to the practical consequences in Section 5.

3.2 ENCODER ARCHITECTURE

The encoder is a multi-layer transformer (Vaswani et al., 2023) adapted to these fixed-length chess tokens. We evaluate two capacity points, SOLIS Mini and SOLIS Base, so that performance can be compared across model size as well as search depth and width. Table 1 reports the architectural quantities used in all experiments.

Table 1: Transformer configurations compared in the experiments, showing the Mini and Base capacity settings used for all reported search-depth and search-width evaluations.

Model	Layers	Embedding dim. D	Heads	MLP Size	Params
SOLIS Mini	6	128	8	256	0.8M
SOLIS Base	6	1024	16	1024	41M

Each token sequence is mapped to D -dimensional vectors with a learned embedding matrix. A classification token (CLS) (Devlin et al., 2019; Dosovitskiy et al., 2021) is prepended so the network can aggregate information across all 77 positions, and learned positional encodings are added because the sequences have fixed length (Ruoss et al., 2024). Stacked transformer encoder layers with GELU (Hendrycks and Gimpel, 2023) activations and dropout (Srivastava et al., 2014) process the sequence. The final CLS hidden state is projected linearly and ℓ_2 normalized, producing the embedding on which all downstream similarity and advantage comparisons are made.

3.3 SUPERVISED CONTRASTIVE TRAINING

The encoder is trained on 5 million randomly sampled ChessBench positions (Ruoss et al., 2024), each paired with a Stockfish-evaluated win probability for the player to move. We normalize every target to White’s perspective, so the quantitative scale has a consistent interpretation: values near 1.0 indicate decisive White advantage, and values near 0.0 indicate decisive Black advantage.

Positive pairs are defined by an explicit evaluation margin. For each anchor, δ sets the maximum win-probability difference for two positions to count as similarly evaluated; we precompute all positions within this margin and sample five positives during training. The batch mask applies the same criterion by comparing every pair of win probabilities and marking pairs positive when their absolute difference is below δ . This construction makes the training signal directly quantitative: positions with nearly equal evaluations are pulled together, while positions separated by larger evaluation gaps are pushed apart, encouraging an ordered latent transition from losing to winning regions.

We optimize the encoder with supervised contrastive learning (SupCon) (Khosla et al., 2021) using these masks as the source of positive and negative comparisons. Cosine similarity supplies the pairwise score, and temperature $\tau = 0.07$ (Girdhar et al., 2023; Radford et al., 2021) controls how sharply the loss distinguishes close from distant examples. Optimization uses stochastic gradient descent with momentum 0.9 (Dosovitskiy et al., 2021; Chen et al., 2020; He et al., 2015), batch size

128, and 400,000 steps on six NVIDIA L40S GPUs (48 GB each) with data parallelism (Paszke et al., 2019). Appendix A.2 lists the complete hyperparameter setting used to reproduce these comparisons.

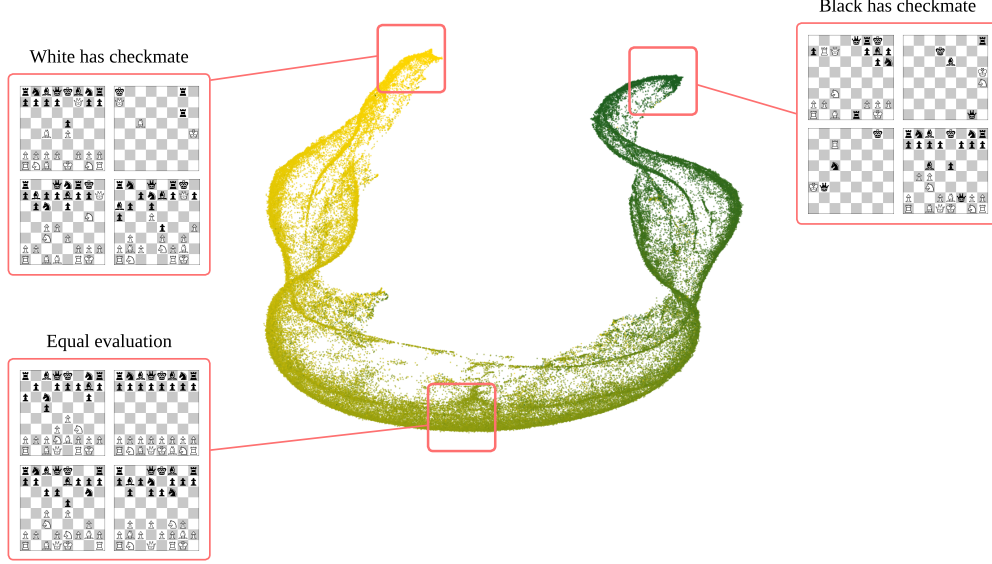


Figure 1: UMAP projection of SOLIS Base embeddings, colored by Stockfish win probability from White’s perspective; gold indicates White-favored positions and green indicates Black-favored positions, making the learned evaluation gradient visible.

Figure 1 projects the SOLIS Base representation with UMAP (McInnes et al., 2020) to show how the training comparisons organize held-out positions. Each point is an encoded test position colored by Stockfish win probability from White’s perspective. The color gradient separates strong White outcomes in gold (*White has checkmate*) from strong Black outcomes in dark green (*Black has checkmate*), illustrating that the learned space preserves the quantitative ordering used for planning.

3.4 EMBEDDING-GUIDED SEARCH

At inference, SOLIS converts the learned geometry into a move ranking by comparing legal continuations against one global advantage direction. The direction is estimated from held-out positions at the two evaluation extremes,

$$\mu_{\text{White}} = \mathbb{E}[z \mid p=1.0], \quad \mu_{\text{Black}} = \mathbb{E}[z \mid p=0.0],$$

and define the normalized advantage vector

$$\vec{a} = \frac{\mu_{\text{White}} - \mu_{\text{Black}}}{\|\mu_{\text{White}} - \mu_{\text{Black}}\|}.$$

We evaluate two scalar scoring rules for this vector so that the effect of the comparison origin can be measured directly.

3.4.1 UNANCHORED (GLOBAL-DIRECTIONAL)

The unanchored rule scores each child embedding by direct alignment with the advantage direction. For child embedding z' , the score is

$$\text{score}_{\text{dir}}(z') = \langle z', \vec{a} \rangle,$$

so higher-ranked moves are those whose embeddings lie furthest along the advantage direction, independent of the current origin. Figure 2 illustrates this direct-similarity scoring rule.

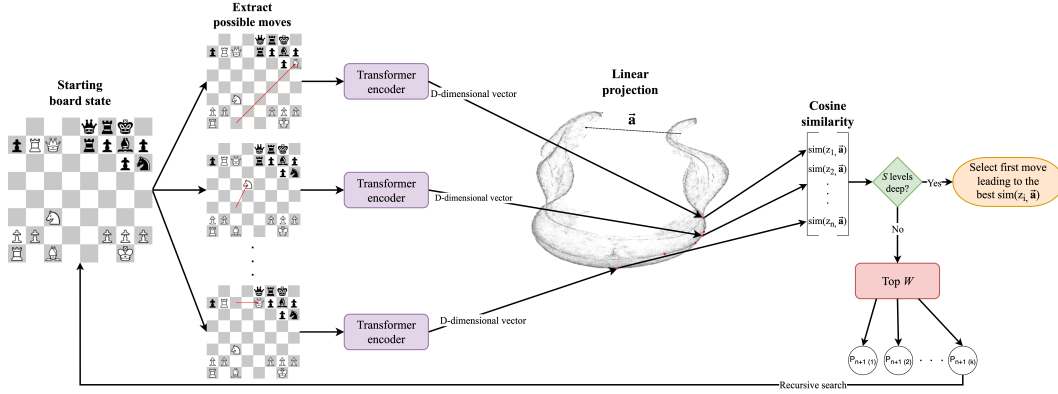


Figure 2: System overview of SOLIS search. Candidate moves are embedded, compared by alignment with the global advantage vector, pruned to the top- W continuations, and recursively evaluated to depth S ; the diagram shows the unanchored score $\cos(z', \vec{a})$ for simplicity.

3.4.2 ANCHORED (RELATIVE PROJECTION)

The anchored rule instead measures progress relative to the Black extreme by shifting each child embedding before projection:

$$\text{score}_{\text{anch}}(z') = \langle z' - \mu_{\text{Black}}, \vec{a} \rangle,$$

so the scalar value is the signed distance from μ_{Black} along $\vec{a}$. This removes dependence on an arbitrary coordinate origin and, in the experiments below, yields a more stable quantitative basis for comparing continuations.

Both scoring mechanisms use the same bounded search procedure shown in Figure 2. Starting from “Starting board state”, SOLIS “Extract possible moves”, embeds each child in parallel with the “Transformer encoder”, computes either $\text{score}_{\text{dir}}$ or $\text{score}_{\text{anch}}$ in the “Cosine similarity” block, and keeps only the “Top- W ” children. If “ S levels deep?” is reached, the selected line returns the “Select first move...”; otherwise, “Recursive search” continues with min-max preference, where White maximizes and Black minimizes the score. Zobrist hashing (Zobrist, 1990) provides a transposition table so repeated positions are not evaluated redundantly.

4 RESULTS

4.1 RATING CALCULATION

We estimate Elo by comparing each SOLIS configuration against Stockfish under a fixed 50 ms per-move limit (Ruoss et al., 2024); Appendix A.3 gives the exact Stockfish settings. The ablation varies both search depth and width, and each depth/width configuration plays 600+ alternating-color games against Stockfish at multiple Elo caps. We evaluate both unanchored and anchored planning with the same protocol. The Stockfish caps are chosen so every SOLIS configuration has at least one positive-win-rate matchup and one negative-win-rate matchup, giving BayesElo a bracket around the rating estimate. Elo ratings and 95% CIs are then estimated with Bayesian logistic regression in BayesElo (Coulom, 2008) using the default confidence parameter of 0.5. Figure 3 reports unanchored estimates, Figure 4 reports anchored estimates, and Appendix A.6 provides the underlying match tables for reproducibility.

Unanchored ratings Under unanchored planning, the rating curves show a consistent scaling pattern with search depth. Depth supplies the largest gains, while width produces a clear improvement when moving from $W \in \{1, 2\}$ to $W=3$ and then largely plateaus at $W=5$. This margin pattern suggests that the most useful continuations are usually already contained in the top three latent candidates. At shallow depths, SOLIS Base leads Mini by roughly 100–250 Elo; at depths 4 and 5, Mini closes the gap and, for $W \in \{3, 5\}$ under this scoring rule, exceeds Base.

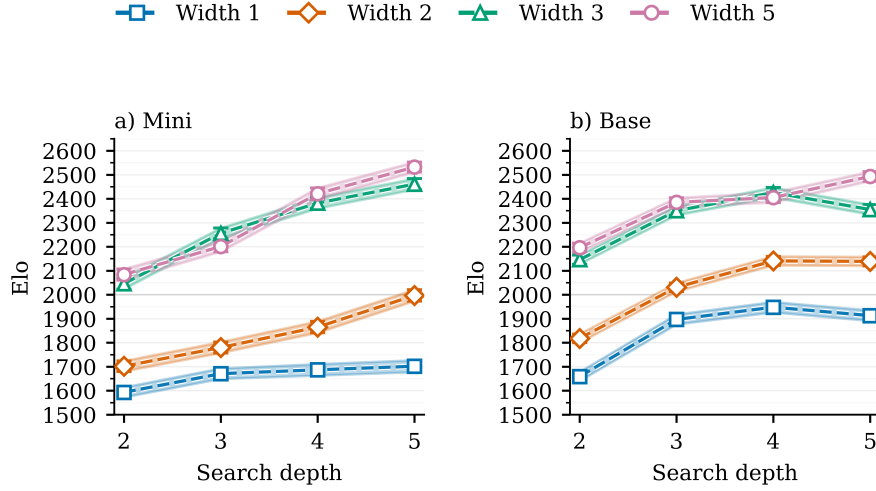


Figure 3: Unanchored Elo estimates for SOLIS Mini and Base across search depths and widths. The curves foreground how rating changes with depth and width under direct similarity scoring; shaded bands denote 95% confidence intervals.

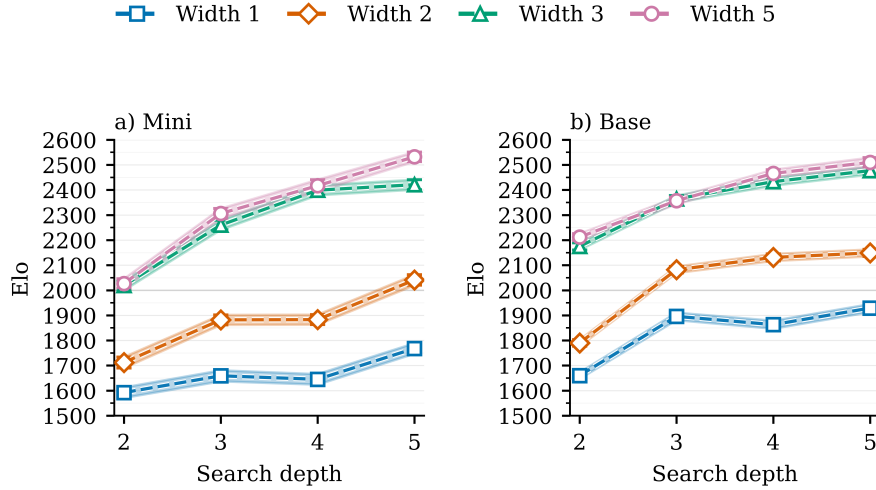


Figure 4: Anchored Elo estimates for SOLIS Mini and Base across search depths and widths. Anchored projection is compared under the same evaluation protocol as unanchored scoring; shaded bands denote 95% confidence intervals.

Anchored ratings Anchored scoring yields a smoother and more consistently ordered set of Elo estimates. Ratings increase with depth and width, the Base–Mini margin remains roughly 100–250 Elo, and both models exceed 2500 Elo at depth 5. Unlike the unanchored case, Base remains stronger than Mini across all reported anchored settings. For Base, anchoring adds a steady 10–30 Elo over the corresponding unanchored configuration, indicating that measuring projection relative to the Black mean improves the reliability of move comparisons.

4.2 EQUAL-DEPTH SIMULATIONS

We next compare SOLIS and Stockfish directly at matched nominal search depths. Following the Elo ablation in Section 4.1, SOLIS uses $W=3$, because increasing width beyond three produced only marginal rating gains. This fixes SOLIS to a small and interpretable comparison budget, whereas

Stockfish uses its default expansion policy with no fixed branching factor. For each depth $S \in \{3, 4, 5\}$, we simulate 200 alternating-color games and report both unanchored and anchored scoring in Table 2.

Simulation results With unanchored scoring, SOLIS Base scores above Stockfish at all tested depths: 106–94 at depth 3, 109–91 at depth 4, and 107.5–92.5 at depth 5. Mini is also competitive, winning at depth 3 (108–92), nearly tying at depth 5 (101–99), and trailing at depth 4 (85–115). With anchored scoring, Base is narrowly behind at depth 3 (98–102) but regains clear margins at depths 4 and 5 (108–92 and 110–90), while Mini wins at depths 3 and 5 (105–95 and 113–87) and trails at depth 4 (91–109).

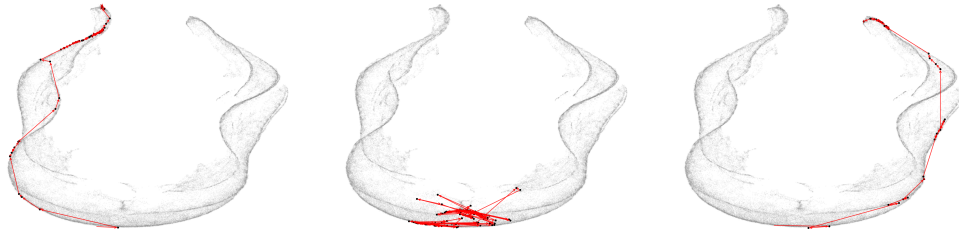
Table 2: Equal-depth match points for SOLIS versus Stockfish over 200 games per row. The middle columns compare direct similarity with the global advantage vector, and the right columns compare anchored projection; higher score is better, and draws count as 0.5.

Config	Depth	Direct Similarity		Anchored Projection	
		SOLIS score	Stockfish score	SOLIS score	Stockfish score
Base	3	106	94	98	102
	4	109	91	108	92
	5	107.5	92.5	110	90
Mini	3	108	92	105	95
	4	85	115	91	109
	5	101	99	113	87

Nodes searched The equal-depth results are especially favorable when normalized by search effort. With fixed branching factor $W=3$, SOLIS searches at most $N(S) = \sum_{i=0}^S 3^i$, which gives $N(3) = 40$, $N(4) = 121$, and $N(5) = 364$. Stockfish’s node count varies with pruning and heuristics: at depth 3 it averages 137 nodes with maximum 1625, at depth 4 it averages 214 nodes with maximum 4463, and at depth 5 it averages 338 nodes with maximum 10611. Thus SOLIS remains competitive while using a controlled node budget whose maximum is far below Stockfish’s observed maxima, supporting the interpretation that **much of the strength normally gained from broad expansion can be recovered with higher-quality representations**.

4.3 INTERPRETABLE GAME TRAJECTORIES

We also examine whether the quantitative ordering in the embedding produces readable trajectories during play. Figure 5 compares three representative games: a White win, a balanced draw, and a Black win. Each position is embedded independently, plotted on the same 2D projection, and connected with arrows to show temporal progression. The empirical pattern matches the evaluation framing: decisive games move smoothly toward the relevant winning region, whereas the drawn game fluctuates around the neutral center. These trajectories complement Figure 1 by showing that the same latent axis used for move comparison also organizes full-game dynamics.



(a) White-win trajectory moves toward the White-favored region. (b) Draw trajectory remains near the neutral region. (c) Black-win trajectory moves toward the Black-favored region.

Figure 5: Latent trajectories for three representative games in the shared representation space. Red arrows show move-by-move progression, highlighting the empirical pattern that wins move toward favored regions while draws remain comparatively central.

5 DISCUSSION

5.1 LIMITATIONS

Blindness to threefold repetition The same lightweight scoring rule that gives SOLIS its favorable node-count profile also removes historical information from the comparison. Because moves are selected by alignment with the advantage vector, SOLIS does not explicitly track whether a position has occurred before. This matters for threefold repetition: if the same position, including side to move, castling rights, and en-passant rights, occurs three times, either player can claim a draw. An opponent can therefore use repetition, including perpetual checks, to reduce an otherwise favorable evaluation to a draw. Game *C* in Appendix A.4 illustrates this failure mode: SOLIS reaches a winning position against Stockfish but allows the practical result to fall to a premature draw by repetition.

Indecisiveness in winning positions A second limitation is that the evaluation target saturates at winning outcomes and therefore does not rank forced wins by length. Mate-in-1 and mate-in-20 both have win probability 1.0, even though the shorter line is practically superior. When SOLIS chooses a longer winning continuation, it preserves the quantitative label but may give the opponent more opportunities to complicate the position, leading in some cases to unnecessary draws or losses. Ruoss et al. (2024) observed the same issue and avoided it by consulting Stockfish as an external oracle. SOLIS instead follows its own latent comparison rule, leaving this as a concrete calibration gap. Game *I* in Appendix A.5 shows a case where many consecutive continuations are winning, but suboptimal choices eventually lead to a draw.

5.2 CONCLUSION

We evaluated whether an evaluation-aligned latent space can carry enough quantitative structure to support planning without a policy head, value head, or learned dynamics model. SOLIS learns a contrastive embedding in which proximity reflects outcome similarity and a single global *advantage vector* orders positions from losing to winning regions. Inference then reduces to a small set of vector comparisons: candidate actions are ranked by alignment with this direction and expanded only within a narrow search budget.

The chess results make the empirical pattern clear. With fixed branching factor $W=3$ and shallow depths $S \in \{3, 4, 5\}$, SOLIS matches or exceeds the configured Stockfish baseline in several equal-depth matchups while using a controlled 40–364 node budget, compared with Stockfish averages in the hundreds and maxima in the thousands. Elo increases with depth; under anchored scoring, both Mini and Base exceed 2500 Elo at depth five, and Base gains a consistent 10–30 Elo from anchoring. Width contributes little beyond three candidates, suggesting that the main practical strength comes from the learned representation rather than broad expansion. The trajectory analyses show the same structure qualitatively: decisive games move toward winning regions, whereas draws remain closer to neutral.

Overall, the measured margins support evaluation-aligned latent planning as a simple and compute-efficient alternative to rollout-heavy search or explicit dynamics modeling. The approach should be most useful in domains where reliable state evaluations can be obtained and where reducing branching factor has practical value. Future work should test this comparison-driven framing beyond chess, add rule-aware state features, prefer shorter forced wins, combine SOLIS with policy priors or MCTS, and study calibration and uncertainty.

REPRODUCIBILITY STATEMENT

Appendix A.2 reports the full training setup, including hyperparameters and implementation details. Section 3 defines the encoder architecture, input representation, supervised contrastive objective, and inference rule, with the compared model configurations summarized in Table 1. Section 4 and Appendix A.3 specify the evaluation protocol, Stockfish settings, rating calculations, and match conditions. Appendix A.6 provides the detailed match outcomes used to verify the Elo estimates.

REFERENCES

- T. Chen, S. Kornblith, M. Norouzi, and G. Hinton. A simple framework for contrastive learning of visual representations, 2020. URL <https://arxiv.org/abs/2002.05709>.
- S. Chowdhury, G. Hamerly, and M. McGarrity. Active learning strategy using contrastive learning and k-means for aquatic invasive species recognition. In *2024 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pages 848–858, 2024. doi: 10.1109/WACVW60836.2024.00097.
- R. Coulom. Whole-history rating: A bayesian rating system for players of time-varying strength. In *Computers and Games*, volume 5131 of *LNCS*, pages 113–124. Springer, 2008. doi: 10.1007/978-3-540-87608-3_10.
- J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding, 2019. URL <https://arxiv.org/abs/1810.04805>.
- A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021. URL <https://arxiv.org/abs/2010.11929>.
- R. Gieslmann and F. T. Pokorny. Latent planning via expansive tree search. In A. H. Oh, A. Agarwal, D. Belgrave, and K. Cho, editors, *Advances in Neural Information Processing Systems*, 2022. URL <https://openreview.net/forum?id=zSdz5scsnzU>.
- R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra. Imagebind: One embedding space to bind them all, 2023. URL <https://arxiv.org/abs/2305.05665>.
- D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, and J. Davidson. Learning latent dynamics for planning from pixels, 2019. URL <https://arxiv.org/abs/1811.04551>.
- D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi. Dream to control: Learning behaviors by latent imagination, 2020. URL <https://arxiv.org/abs/1912.01603>.
- D. Hafner, T. Lillicrap, M. Norouzi, and J. Ba. Mastering atari with discrete world models, 2022. URL <https://arxiv.org/abs/2010.02193>.
- D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models, 2024. URL <https://arxiv.org/abs/2301.04104>.
- J. B. Hamrick, V. Bapst, A. Sanchez-Gonzalez, T. Pfaff, T. Weber, L. Buesing, and P. W. Battaglia. Combining q-learning and search with amortized value estimates, 2020. URL <https://arxiv.org/abs/1912.02807>.
- K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition, 2015. URL <https://arxiv.org/abs/1512.03385>.

540 D. Hendrycks and K. Gimpel. Gaussian error linear units (gelus), 2023. URL <https://arxiv.org/abs/1606.08415>.

541

542

543 M. Janner, J. Fu, M. Zhang, and S. Levine. When to trust your model: Model-based policy opti-

544 mization, 2021. URL <https://arxiv.org/abs/1906.08253>.

545

546 Z. Jiang, T. Zhang, M. Janner, Y. Li, T. Rocktäschel, E. Grefenstette, and Y. Tian. Efficient planning

547 in a compact latent action space, 2023. URL <https://arxiv.org/abs/2208.10291>.

548

549 P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan.

550 Supervised contrastive learning, 2021. URL <https://arxiv.org/abs/2004.11362>.

551

552 M. Lai. Giraffe: Using deep reinforcement learning to play chess, 2015. URL <https://arxiv.org/abs/1509.01549>.

553

554 L. McInnes, J. Healy, and J. Melville. Umap: Uniform manifold approximation and projection for

555 dimension reduction, 2020. URL <https://arxiv.org/abs/1802.03426>.

556

557 T. M. Moerland, J. Broekens, A. Plaat, and C. M. Jonker. Model-based reinforcement learning: A

558 survey, 2022. URL <https://arxiv.org/abs/2006.16712>.

559

560 Y. Nasu. Efficiently updatable neural-network-based evaluation functions for computer shogi, 2018.

561 Accessed: 2024-04-25.

562

563 G.-C. Pascutto and G. Linscott. Leelachesszero, 2018. URL <https://lczero.org>. Accessed:

564 2024-04-25.

565

566 A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein,

567 L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy,

568 B. Steiner, L. Fang, J. Bai, and S. Chintala. Pytorch: An imperative style, high-performance deep

569 learning library, 2019. URL <https://arxiv.org/abs/1912.01703>.

570

571 M. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. Wiley

572 Series in Probability and Statistics, Wiley, 2014. ISBN 9781118625873. URL <https://books.google.com/books?id=VvBjBAAQBAJ>.

573

574 A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin,

575 J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language

576 supervision, 2021. URL <https://arxiv.org/abs/2103.00020>.

577

578 T. Romstad, M. Costalba, J. Kiiski, G. Linscott, Y. Nasu, M. Isozaki, and H. Noda. Stockfish, 2008.

579 URL <https://stockfishchess.org>. Accessed: 2025-04-22.

580

581 A. Ruoss, G. Delétang, S. Medapati, J. Grau-Moya, L. K. Wenliang, E. Catt, J. Reid, C. A. Lewis,

582 J. Veness, and T. Genewein. Amortized planning with large-scale transformers: A case study on

583 chess, 2024. URL <https://arxiv.org/abs/2402.04494>.

584

585 S. Russell and P. Norvig. *Artificial Intelligence, A modern approach*. 1995.

586

587 J. Schrittwieser, I. Antonoglou, T. Hubert, K. Simonyan, L. Sifre, S. Schmitt, A. Guez, E. Lock-

588 hart, D. Hassabis, T. Graepel, T. Lillicrap, and D. Silver. Mastering atari, go, chess and

589 shogi by planning with a learned model. *Nature*, 588(7839):604–609, Dec. 2020. ISSN

590 1476-4687. doi: 10.1038/s41586-020-03051-4. URL <http://dx.doi.org/10.1038/s41586-020-03051-4>.

591

592 C. E. Shannon. XXII. programming a computer for playing chess. *The London, Edinburgh, and*

593 *Dublin Philosophical Magazine and Journal of Science*, 41(314):256–275, 1950.

D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Ku-

maran, T. Graepel, T. Lillicrap, K. Simonyan, and D. Hassabis. Mastering chess and shogi by

self-play with a general reinforcement learning algorithm, 2017. URL <https://arxiv.org/abs/1712.01815>.

-
- N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.*, 15(1):1929–1958, Jan. 2014. ISSN 1532-4435.
- R. S. Sutton. Dyna, an integrated architecture for learning, planning, and reacting. *ACM Sigart Bulletin*, 2(4):160–163, 1991.
- R. S. Sutton and A. G. Barto. *Reinforcement learning: An introduction*, volume 1. MIT press Cambridge, 1998.
- J. v. Neumann. Zur theorie der gesellschaftsspiele. *Mathematische annalen*, 100(1):295–320, 1928.
- A. van den Oord, Y. Li, and O. Vinyals. Representation learning with contrastive predictive coding, 2019. URL <https://arxiv.org/abs/1807.03748>.
- A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need, 2023. URL <https://arxiv.org/abs/1706.03762>.
- A. L. Zobrist. A new hashing method with application for game playing. *ICGA Journal*, 13(2): 69–73, 1990.

A APPENDIX

A.1 USE OF LARGE LANGUAGE MODELS (LLMs)

Large language models were used only for minor presentation tasks, including formatting and figure/table placement. They were not used to design the experiments, compute or interpret the quantitative results, or generate technical content. All methodological choices and empirical claims were developed and verified by the authors.

A.2 TRAINING SETUP

Both SOLIS configurations, Mini and Base, are trained with the same quantitative objective and data scale: supervised contrastive learning on 5M ChessBench positions, cosine similarity over ℓ_2 -normalized embeddings, and SGD with momentum on 6 48 GB L40S GPUs for 400k steps with batch size 128.

Among these settings, τ and δ most directly shape the pairwise comparisons. The temperature $\tau = 0.07$ follows prior contrastive work (Radford et al., 2021; Girdhar et al., 2023): smaller values sharpen similarity differences, while larger values smooth gradients. We trained with $\tau \in \{0.05, 0.07, 0.1\}$ and found that 0.07 converged most reliably across model sizes. The margin $\delta = 0.05$ sets the evaluation tolerance for positives, balancing overly broad neighborhoods when δ is large against excessive fragmentation when δ is small; it also reflects Stockfish evaluation noise, since sub-0.05 differences are rarely meaningful in practice.

A.3 STOCKFISH SETUP

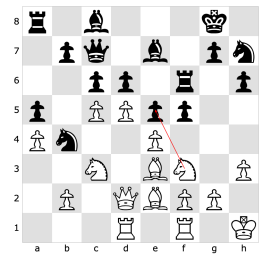
All Stockfish comparisons use the official Linux release of Stockfish 16 (Romstad et al., 2008) with 32 MB hash and default threading. Elo-capped baselines are produced with `UCI_LimitStrength` and `UCI_Elo`, while equal-depth baselines use the `depth` parameter. For node-count comparisons, we set the engine information level to `INFO_ALL` and record the reported `nodes` attribute for each move.

Table 3: Training hyperparameters shared by SOLIS Mini and Base for the supervised-contrastive embedding experiments.

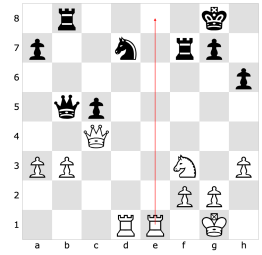
Attribute	Value
Loss	SupCon
τ	0.07
δ (positive margin)	0.05
Positives per anchor	5
Optimizer	SGD
Momentum	0.9
Learning rate	0.05
Batch size	128
Steps	400k
Dropout	0.10
Activation	GELU
Similarity	cosine
Embedding norm	ℓ_2
Token length	77

A.4 REPRESENTATIVE GAMES: STOCKFISH (CAP) VS. SOLIS (DEPTH, WIDTH)

The following examples show individual comparison outcomes between SOLIS and rating-capped Stockfish. SOLIS is limited to search depth S and width W , while Stockfish uses unbounded search subject to its Elo cap. Games *A*, *B*, and *E* illustrate favorable tactical and long-horizon conversions by SOLIS; Games *C* and *D* illustrate limitations that are consistent with the aggregate discussion. In each table, the left column gives the PGN move sequence and the right column shows a critical position.

1. d4 Nf6 2. c4 d6 3. Nc3 e5 4. Nf3 Qe7 5. e4 c6 6. d5 Qc7 7. Be2 h6 8. a4 a5 9. O-O Be7 10. h3 Na6 11. Be3 Nb4 12. Kh1 Nh7 13. Qd2 O-O 14. Rad1 f5 15. c5 Rf6 16. Nxe5 f4 17. Bxf4 Rxf4 18. Qxf4 dxe5 19. d6 exf4 20. Bc4+ Kf8 21. dxc7 Ke8 22. e5 Ng5 23. Rd6 Bxd6 24. exd6 Bd7 25. Re1+ Kf8 26. Re7 Bf5 27. h4 Nd5 28. Nxd5 cxd5 29. Bxd5 Be6 30. Bxe6 f3 31. hxg5 b6 32. Bh3 hxg5 33. Re5 Rc8 34. Rf5+ Kg8 35. Rf7 g4 36. cxb6 g6 37. d7 fxg2+ 38. Kh2 Rf8 39. Bxg2 Kxf7 40. Bc6 Ke7 41. Kg3 Rf3+ 42. Kh2 Rd3 43. b7 Rh3+ 44. Kg2 Re3 45. d8=Q+ Kf7 46. b8=Q Rg3+ 47. Kh2 Rh3+ 48. Kg1 Rh7 49. Qe8+ Kf6 50. Qbd8+ Kf5 51. Be4+ Kf4 52. Qf6# 1-0	
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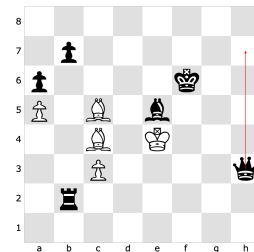
Game A — SOLIS ($S=5$, $W=3$) vs. SF 2350, SOLIS–White. Result: 1–0. SOLIS converts the 2350-capped Stockfish comparison by sacrificing a knight and turning the resulting middlegame advantage into a win.

1. d4 d5 2. e3 c6 3. Bd3 h6 4. h3 Nf6 5. Nf3 e6 6. O-O b6 7. c4 Ba6 8. b3 Nbd7 9. Bb2 Bb4 10. a3 Be7 11. Nbd2 O-O 12. Re1 c5 13. Qe2 Qc7 14. e4 Nxe4 15. Nxe4 dxe4 16. Bxe4 Rab8 17. d5 Bf6 18. dxe6 Bxb2 19. exf7+ Rxf7 20. Qxb2 Bb7 21. Bxb7 Qxb7 22. Rad1 b5 23. Qe2 bxc4 24. Qxc4 Qb5 25. Re8+ Nf8 26. Rxb8 Qxb8 27. Rd5 Qf4 28. Qxf4 Rxf4 29. Rxc5 Ne6 30. Rc6 Nd4 31. Nxd4 a5 32. Rc4 Rf8 33. Nc6 Kh8 34. Nxa5 Ra8 35. b4 Rd8 36. g3 Rd3 37. a4 g5 38. b5 Rd2 39. b6 Rd5 40. b7 Rxa5 41. Rc8+ Kh7 42. b8=Q Rd5 43. Rc7+ Kg6 44. Qg8+ Kf5 45. Qxd5+ Kg6 46. Qf7# 1-0	
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Game B — SOLIS ($S=5$, $W=3$) vs. SF 2300, SOLIS–White. Result: 1–0. Against the 2300-capped Stockfish baseline, SOLIS sacrifices a rook, wins the queen, and preserves the material margin through conversion.

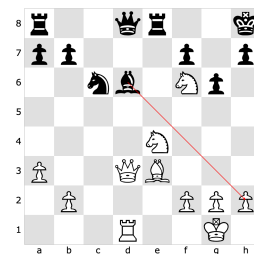
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1. e3 Nf6 2. Nc3 d5 3. Nf3 c5 4. d4 e6 5. g3 a6 6. a4 Nc6 7. Ne2 Bd6 8. Bg2 O-O 9. h3 Re8 10. c3 Qc7 11. Nd2 e5 12. Nb3 c4 13. dxe5 Nxe5 14. Nbd4 Nd3+ 15. Kf1 Nc5 16. Bf3 Bd7 17. Kg2 Be5 18. a5 Rad8 19. Qc2 g6 20. Bd2 h5 21. Be1 h4 22. gxh4 Kg7 23. Ng3 Rh8 24. h5 Rdg8 25. hxg6 fxg6 26. Be2 Qd6 27. Qd1 Qe7 28. f3 Bd6 29. Nf1 Rh5 30. h4 Rgh8 31. Bf2 R5h7 32. Qe1 Bb8 33. Bg3 Ba7 34. Bf2 Nh5 35. Ng3 Bb8 36. f4 Nf6 37. h5 Rh6 38. hxg6 Rxh1 39. Qxh1 Qe8 40. Ndf5+ Bxf5 41. Nxf5+ Kxg6 42. Ne7+ Kf7 43. e4 Rxh1 44. Rxh1 Qxe7 45. Bxc5 Qxe4+ 46. Bf3 Qg6+ 47. Kf1 Bxf4 48. b4 cxb3 49. Rh8 Ke6 50. Rf8 b2 51. Rxf6+ Kxf6 52. Bxd5 b1=R+ 53. Ke2 Rb2+ 54. Kf3 Be5 55. Bc4 Qh5+ 56. Ke3 Qh3+ 57. Ke4 Qh7+ 58. Kf3 Qh3+ 59. Ke4 Qh7+ 60. Kf3 Qh3+ 61. Ke4 Qh7+ 62. Kf3 Qh3+ 63. Ke4 Qh7+ 64. Kf3 Qh3+ 65. Ke4 1/2-1/2



Game C — SF 2200 vs. SOLIS ($S=5, W=3$), SOLIS-Black. Result: 1/2-1/2. SOLIS reaches a winning endgame against the 2200-capped baseline, but the practical score drops to a draw because the search rule does not penalize threefold repetition.

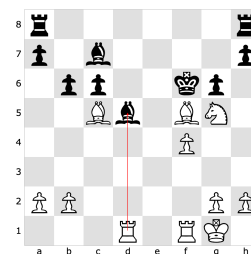
1. d4 d5 2. c4 dxc4 3. e3 e5 4. Bxc4 exd4 5. exd4 Nf6 6. Nc3 Bd6 7. Qe2+ Qe7 8. Be3 O-O 9. Nf3 c6 10. O-O Be6 11. Rfd1 Re8 12. d5 cxd5 13. Bxd5 Nc6 14. a3 Nxd5 15. Nxd5 Qd8 16. Qd3 Bg4 17. Ng5 g6 18. Ne4 Bxd1 19. Ndf6+ Kh8 20. Rxd1 Bxh2+ 21. Kxh2 Qxd3 22. Rxd3 Red8 23. Nd7 Kg7 24. f3 h6 25. Rd6 f5 26. Nec5 Re8 27. Bd2 Re2 28. Kh1 Kf7 29. b3 Ke7 30. Nxb7 Rc8 31. Ndc5 Kf7 32. Kg1 Kg8 33. Rxg6+ Kf7 34. Rd6 Ne7 35. Rd7 Rg8 36. Nd6+ Kg6 37. Nc4 Kh5 38. Ne3 f4 39. Ne6 fxe3 40. Nf4+ Kg5 41. Nxe2 exd2 42. Kf2 Kf6 43. Nc3 Ke6 44. Rd4 Rc8 45. Ne4 d1=N+ 46. Rxd1 Rc2+ 47. Kg1 Nf5 48. Rd2 Rxd2 49. Nxd2 Ke5 50. Kh2 Kf4 51. b4 Ke3 52. Ne4 Nd4 53. a4 Nc2 54. b5 Nd4 55. Nd6 Ne6 56. a5 Kd3 57. g4 Kd4 58. b6 axb6 59. axb6 Nc5 60. Kg3 Kd5 61. b7 Na6 62. Nf7 Kc6 63. Nxb6 Kxb7 64. f4 Kc7 65. Kh4 Kd7 66. Kh5 Nc5 67. f5 Kd6 68. f6 Ne6 69. Nf7+ Kc6 70. Kh6 Kd5 71. g5 Nc5 72. Kg6 Ne6 73. Kh5 Kd4 74. g6 Kc4 75. Ne5+ Kd5 76. Ng4 Ke4 77. Kh6 Kf5 78. g7 Nxg7 79. Kxg7 Kxg4 80. Kf7 Kf5 81. Ke7 Kg5 82. f7 Kf4 83. Kf6 Kf3 84. Ke5 Kg3 85. Ke4 Kf2 86. Kd3 Kg1 87. Ke3 Kh2 88. f8=Q Kg3 89. Qg8+ Kh3 90. Kf4 Kh2 91. Kf3 Kh3 92. Qg3# 1-0



Game D — SF 2500 vs. SOLIS ($S=5, W=3$), SOLIS-Black. Result: 1-0 This loss against the 2500-capped baseline highlights a negative case: SOLIS chooses a locally attractive sacrifice whose long-term material cost is not recovered.

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1. d4 d5 2. e3 Nd7 3. c4 e5 4. dxe5 dxc4 5. Nf3
Qe7 6. Bxc4 Nxe5 7. Nxe5 Qxe5 8. Qb3 Qh5 9. e4
Nf6 10. O-O Bd6 11. Qb5+ c6 12. Qxh5 Nxh5 13. Nc3
Be5 14. f4 Bc7 15. Be3 g6 16. Bb3 Be6 17. Rad1 Ng7
18. Bc2 f5 19. Bc5 Kf7 20. exf5 Nxf5 21. Ne4 b6
22. Ng5+ Kf6 23. Bxf5 Bd5 24. Rxd5 cxd5 25. Bd4+
Kxf5 26. h3 Bxf4 27. Bxh8 Kxg5 28. Bg7 Rg8 29. Bd4
Re8 30. Rf3 Re1+ 31. Kf2 Re6 32. g3 Bd2 33. Rf7
Re4 34. Bf6+ Kh6 35. g4 a6 36. Kg3 d4 37. h4 Be1+
38. Kh3 Re3+ 39. Kg2 Bxh4 40. Bxh4 a5 41. Bf6 a4
42. Rd7 Rd3 43. Rd5 b5 44. Rxb5 a3 45. Re5 Rd2+
46. Kf3 axb2 47. g5+ Kh5 48. Rb5 Rd3+ 49. Kf4 h6
50. gxh6+ Kxh6 51. Rxb2 g5+ 52. Kg4 Rd1 53. Bxg5+
Kg6 54. Rb6+ Kg7 55. Kf5 Kf7 56. Rb7+ Ke8 57. a4
Ra1 58. Re7+ Kf8 59. Rh7 Rxa4 60. Bh6+ Kg8 61. Re7
Rc4 62. Kg5 Rc5+ 63. Kg4 Ra5 64. Rg7+ Kh8 65. Re7
d3 66. Rd7 d2 67. Bxd2 Rc5 68. Bf4 Kg8 69. Be3 Rb5
70. Bh6 Re5 71. Kf4 Rc5 72. Ke4 Rb5 73. Kf4 Rb4+
74. Ke5 Rb1 75. Kf6 Rb2 76. Kg6 Rb6+ 77. Kh5 Rb2
78. Bg5 Rb6 79. Bh6 Re6 80. Kg5 Ra6 81. Rg7+ Kh8
82. Re7 Ra5+ 83. Kg6 Ra6+ 84. Kh5 Kg8 85. Rb7 Rc6
86. Rd7 Rc5+ 87. Bg5 Rc2 88. Kg6 Rc6+ 89. Bf6 Re6
90. Rd8+ Re8 91. Rxe8# 1-0

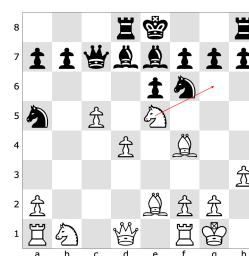


Game E — SOLIS ($S=5$, $W=3$) vs. Stockfish 2500 , SOLIS-White. Result: 1-0 SOLIS defeats the 2500-capped baseline by making an exchange sacrifice for long-term compensation and maintaining enough advantage to convert.

A.5 EQUAL-DEPTH MATCHES: SOLIS D vs. STOCKFISH D

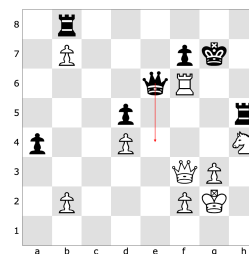
We include several illustrative games where SOLIS and Stockfish search to the same depth D . These examples emphasize how SOLIS' latent planning competes head-to-head without a rating cap on Stockfish. The left portion of each table encodes the moves of the game in Portable Game Notation (PGN) format; the right portion of each table displays a critical moment in the game.

1. d4 d5 2. e3 c5 3. Nf3 Nf6 4. Be2 e6 5. O-O Nc6
6. b3 cxd4 7. exd4 Be7 8. c3 Qc7 9. h3 Bd7 10. c4
dxc4 11. bxc4 Na5 12. Ne5 Rd8 13. Bf4 Bd6 14. c5
Be7 15. Ng6 Qc8 16. Nxh8 Bc6 17. Nc3 Kf8 18. Rc1
Qa8 19. Bc7 b5 20. Bxa5 Bxg2 21. d5 Bxh3 22. Qd3
Bf5 23. Qg3 b4 24. Rfd1 Bxc5 25. Qc7 Be7 26. Nb5
Nxd5 27. Rxd5 Qxd5 28. Bxb4 Qd7 29. Qxd7 Re8 30.
Rc8 g5 31. Qxe7+ Kg7 32. Qxf7+ Kh6 33. Bf8+ Rxf8 34.
Qxf8# 1-0



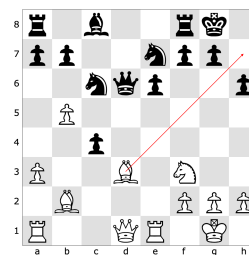
Game F — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 At matched depth $S = 5$, SOLIS sacrifices a knight and converts the resulting tactical margin against Stockfish.

1. d4 d5 2. e3 c5 3. Nf3 e6 4. Be2 cxd4 5. exd4
Bd6 6. O-O Nf6 7. Bg5 Nbd7 8. Nbd2 h6 9. Bh4 O-O
10. Re1 a5 11. Bf1 Bf4 12. c3 Qb6 13. Qc2 Qc7 14.
a4 b6 15. Bb5 Ba6 16. Bxa6 Rxa6 17. Qd3 Ra7 18. Re2
Qc8 19. Reel Nh5 20. Nf1 Nh6 21. N1d2 Nh5 22. Nf1
Bc7 23. Qd1 Nh6 24. Qd3 Nh5 25. Qd1 Nh6 26. Qd3
Rb7 27. Qb5 Ra7 28. N1d2 Nh5 29. h3 g5 30. Rac1
gxh4 31. c4 Nf4 32. Rc3 Qd8 33. g3 Nxh3+ 34. Kg2
Ng5 35. cxd5 h3+ 36. Kg1 h2+ 37. Kxh2 exd5 38. Rc6
Nf6 39. Nxg5 hxg5 40. Kg2 Re8 41. Rh1 Bd6 42. Nf3
Kg7 43. Nxg5 Rc7 44. Rxd6 Qxd6 45. Qf1 b5 46. axb5
Rcc8 47. Rh4 Rh8 48. Nf3 Ne4 49. Rf4 Rh5 50. Qe2
Qe6 51. Nh4 Rch8 52. b6 Nf6 53. Qf3 a4 54. b7 Rb8
55. Rxf6 Qe4 56. Rxf7+ Kh8 57. Qxe4 dxe4 58. Rc7
Rg5 59. Rc8+ Rg8 60. Rc7 Rg5 61. Rc8+ Rg8 62. Ng6+
Kg7 63. Ne7 Rf8 64. Rc7 Rf7 65. d5 Rbf8 66. Nf5+
Kg6 67. Rxf7 Kxf7 68. Nd6+ Kg6 69. Nc8 Kg5 70. b8=Q
Kg4 71. Qc7 Rf5 72. Nd6 e3 73. Qg7+ Rg5 74. f3+ Kh5
75. Qh7# 1-0



Game G — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 At matched depth $S = 5$, Stockfish first builds an advantage, but SOLIS exploits a late-middlegame error and converts the reversal in the endgame.

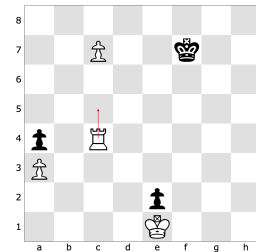
1. d4 c6 2. e4 d5 3. Nc3 Nf6 4. e5 Ng8 5. Bd3 e6
6. Nf3 c5 7. dxc5 Bxc5 8. O-O Nc6 9. a3 h6 10. b4
Bb6 11. Nb5 Bc7 12. Re1 Bb8 13. c4 Nge7 14. Nd6+
Bxd6 15. exd6 Qxd6 16. Bb2 O-O 17. b5 dxc4 18. Bh7+
Kh8 19. Qxd6 f6 20. Be4 c3 21. Bxc3 e5 22. bxc6 Bf5
23. Qxe7 Bxe4 24. Rxe4 bxc6 25. Nxe5 Rfe8 26. Nf7+
Kg8 27. Nxh6+ Kh7 28. Rae1 Rad8 29. Ba5 Rd4 30. g4
Rxe4 31. Rxe4 Rb8 32. Qxa7 Rd8 33. Rf4 Rd3 34. g5
fxg5 35. Rd4 Rxa3 36. Kf1 Rb3 37. Rb4 Rc3 38. h4
gxh4 39. Rd4 h3 40. Qe7 h2 41. Qe4+ Kxh6 42. Rd6+
Kg5 43. Rg6+ Kh5 44. Qg4# 1-0



Game H — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1-0 At matched depth $S = 3$, SOLIS sacrifices a bishop to create a discovered attack on Stockfish's queen and converts the resulting advantage.

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1. d4 d5 2. e3 c6 3. c4 e6 4. Nf3 Nf6 5. Be2 c5
6. O-O Nc6 7. b3 a6 8. Bb2 Be7 9. a3 cxd4 10. exd4
O-O 11. c5 b6 12. b4 bxc5 13. dxc5 Bb7 14. Nbd2 a5
15. b5 Bxc5 16. Rc1 Qb6 17. bxc6 Ng4 18. Rxc5 Ba6
19. Rc2 Nxf2 20. Rxf2 Bxe2 21. Qxe2 a4 22. c7 Rac8
23. Bd4 Qb5 24. Qxb5 Rfe8 25. Qb8 h6 26. Ne5 g5 27.
h4 Rf8 28. Ne4 dxe4 29. Rxf7 Rxb8 30. Rxf8+ Rxf8 31.
hxcg5 Rc8 32. gxh6 e3 33. Bxe3 Kh7 34. Rc4 Kg8 35.
Kh2 Kh7 36. g4 Rg8 37. g5 Ra8 38. Rg4 Kh8 39. Ba7
Rf8 40. Bb8 Rf2+ 41. Kg1 Rc2 42. Rc4 Rb2 43. Ba7
Rb1+ 44. Kg2 Kh7 45. Bf2 Rb2 46. g6+ Kxh6 47. Kh2
Rxf2+ 48. Kg3 Rg2+ 49. Kxg2 Kg7 50. Rc1 Kf6 51. Kh2
Kg7 52. Nd7 e5 53. Nf6 Kxf6 54. Rg1 Kg7 55. Rf1 Kh6
56. Rg1 Kg7 57. Rf1 Kxg6 58. Rc1 Kg7 59. Rc4 Kf7
60. Kg2 Ke7 61. Kf2 Kd6 62. Ke1 Kd5 63. Rc1 Ke4 64.
Ke2 Kd4 65. Rc2 e4 66. Rc1 Kd5 67. Ke1 Ke5 68. Rc4
Kd5 69. Rc2 Kd6 70. Rc1 Ke7 71. Rc4 Kf7 72. Rc5 e3
73. Kf1 Ke7 74. Rc4 Kf7 75. Kg2 e2 76. Kf2 Ke7 77.
Ke1 Kf7 78. Rc5 Kg7 79. Rc2 Kf6 80. Rc4 Kf7 81. Rc5
Kg6 82. Kd2 Kf7 83. Ke1 Ke7 84. Rc4 Kf7 85. Rc5 Kg6
86. Rc4 Kf7 87. Rc5 1/2-1/2



Game I — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1/2-1/2 This drawn matched-depth game illustrates saturation: many continuations are winning by evaluation, but SOLIS does not reliably rank the fastest conversion.

A.6 SOLIS VS. STOCKFISH RESULTS

To make the Elo estimates auditable, we report the full SOLIS–Stockfish match outcomes for SOLIS depths $S \in \{2, 3, 4, 5\}$. Each table indexes the comparison by Stockfish Elo cap in rows and SOLIS search width W in columns. Entries give win–loss–draw counts for the matchup, with total match points in parentheses. Separate tables distinguish Base versus Mini and unanchored versus anchored scoring, preserving the exact empirical inputs used for rating estimation.

Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–

Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 6: Full Base-model comparison against Stockfish under unanchored planning, grouped by SOLIS depth and search width. Each cell reports W–D–L and total match score out of 100, enabling direct inspection of the match margins used for Elo estimation.

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Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	32–32–36 (48)	27–45–28 (50)	–	–	1350	–	–	–	–
1400	30–41–29 (50)	33–27–40 (46)	–	–	1400	–	–	–	–
1450	29–44–27 (51)	33–40–27 (53)	–	–	1450	–	–	–	–
1500	31–38–31 (50)	31–39–30 (50)	–	–	1500	29–38–33 (48)	28–43–29 (50)	–	–
1550	38–35–27 (56)	23–50–27 (48)	–	–	1550	41–36–23 (59)	36–37–27 (54)	–	–
1600	34–30–36 (49)	29–37–34 (48)	–	–	1600	37–30–33 (52)	37–30–33 (52)	–	–
1650	24–45–31 (46)	28–36–36 (46)	–	–	1650	32–38–30 (51)	30–40–30 (50)	–	–
1700	–	–	–	–	1700	28–37–35 (46)	21–43–36 (42)	–	–
1750	–	–	–	–	1750	36–33–31 (52)	28–40–32 (48)	–	–
1800	–	–	37–31–32 (52)	37–36–27 (55)	1800	39–24–37 (51)	29–35–36 (46)	–	–
1850	–	–	34–24–42 (46)	34–31–35 (50)	1850	–	–	–	–
1900	–	–	40–28–32 (54)	40–29–31 (54)	1900	–	–	–	–
1950	–	–	39–29–32 (54)	39–29–32 (54)	1950	–	–	–	–
2000	–	–	38–34–28 (55)	42–28–30 (56)	2000	–	–	34–35–31 (52)	38–21–41 (48)
2050	–	–	41–27–32 (54)	40–26–34 (53)	2050	–	–	38–34–28 (55)	34–33–33 (50)
2100	–	–	35–30–35 (50)	36–26–38 (49)	2100	–	–	33–34–33 (50)	34–29–37 (48)
2150	–	–	–	–	2150	–	–	31–35–34 (48)	38–25–37 (50)
2200	–	–	–	–	2200	–	–	38–18–44 (47)	31–26–43 (44)
2250	–	–	–	–	2250	–	–	36–23–41 (48)	45–26–29 (58)
2300	–	–	–	–	2300	–	–	40–22–38 (51)	34–26–40 (47)
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2500	–	–	–	–	2500	–	–	–	–
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	–	–	–	–	1350	–	–	–	–
1400	–	–	–	–	1400	–	–	–	–
1450	–	–	–	–	1450	–	–	–	–
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	36–42–22 (57)	35–31–34 (50)	–	–	1600	–	–	–	–
1650	29–40–31 (49)	36–33–31 (52)	–	–	1650	–	–	–	–
1700	33–34–33 (50)	36–32–32 (52)	–	–	1700	39–30–31 (54)	34–34–32 (51)	–	–
1750	43–26–31 (56)	28–38–34 (47)	–	–	1750	37–29–34 (52)	32–30–38 (47)	–	–
1800	38–24–38 (50)	30–37–33 (48)	–	–	1800	42–20–38 (52)	33–31–36 (48)	–	–
1850	49–23–28 (60)	29–33–38 (46)	–	–	1850	36–24–40 (48)	43–28–29 (57)	–	–
1900	–	–	–	–	1900	45–26–29 (58)	46–31–23 (62)	–	–
1950	–	–	–	–	1950	40–22–38 (51)	37–35–28 (54)	–	–
2000	–	–	–	–	2000	–	–	–	–
2050	–	–	–	–	2050	–	–	–	–
2100	–	–	–	–	2100	–	–	–	–
2150	–	–	31–31–38 (46)	44–21–35 (54)	2150	–	–	–	–
2200	–	–	35–29–36 (50)	48–27–25 (62)	2200	–	–	27–37–36 (46)	33–38–29 (52)
2250	–	–	37–31–32 (52)	37–28–35 (51)	2250	–	–	30–33–37 (46)	38–31–31 (54)
2300	–	–	27–37–36 (46)	36–34–30 (53)	2300	–	–	38–37–25 (56)	33–39–28 (52)
2350	–	–	22–42–36 (43)	37–35–28 (54)	2350	–	–	32–29–39 (46)	29–36–35 (47)
2400	–	–	36–29–35 (50)	37–26–37 (50)	2400	–	–	30–26–44 (43)	30–32–38 (46)
2500	–	–	–	–	2500	–	–	26–34–40 (43)	43–25–32 (56)

Figure 7: Full Mini-model comparison against Stockfish under unanchored planning, grouped by SOLIS depth and search width. Each cell reports W–D–L and total match score out of 100, making the width and depth patterns behind the Elo estimates explicit.

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Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32-46-22 (55)	41-47-12 (64)	-	-	1500	-	-	-	-
1550	39-38-23 (58)	37-50-13 (62)	-	-	1550	-	-	-	-
1600	32-44-24 (54)	33-51-16 (58)	-	-	1600	-	-	-	-
1650	33-41-26 (54)	29-46-25 (52)	-	-	1650	-	-	-	-
1700	34-38-28 (53)	22-53-25 (48)	-	-	1700	-	-	-	-
1750	37-35-28 (54)	30-49-21 (54)	-	-	1750	-	-	-	-
1800	29-42-29 (50)	36-42-22 (57)	-	-	1800	39-29-32 (54)	37-49-14 (62)	-	-
1850	-	-	-	-	1850	35-32-33 (51)	30-50-20 (55)	-	-
1900	-	-	-	-	1900	37-30-33 (52)	40-41-19 (60)	-	-
1950	-	-	-	-	1950	31-28-41 (45)	37-44-19 (59)	-	-
2000	-	-	34-40-26 (54)	43-31-26 (58)	2000	41-27-32 (54)	33-45-22 (56)	-	-
2050	-	-	46-38-16 (65)	36-38-26 (55)	2050	39-27-34 (52)	31-41-28 (52)	-	-
2100	-	-	34-45-21 (56)	34-31-35 (50)	2100	41-27-32 (54)	36-38-26 (55)	-	-
2150	-	-	46-31-23 (62)	30-43-27 (52)	2150	-	-	-	-
2200	-	-	31-35-34 (48)	35-28-37 (49)	2200	-	-	33-42-25 (54)	30-41-29 (50)
2250	-	-	34-34-32 (51)	40-29-31 (54)	2250	-	-	35-41-24 (56)	33-43-24 (54)
2300	-	-	32-32-36 (48)	37-34-29 (54)	2300	-	-	35-33-32 (52)	36-34-30 (53)
2350	-	-	-	-	2350	-	-	37-37-26 (56)	29-37-34 (48)
2400	-	-	-	-	2400	-	-	34-36-30 (52)	37-38-25 (56)
2450	-	-	-	-	2450	-	-	37-28-35 (51)	43-29-28 (58)
2500	-	-	-	-	2500	-	-	41-30-29 (56)	34-30-36 (49)
2550	-	-	-	-	2550	-	-	-	-
2600	-	-	-	-	2600	-	-	-	-
2650	-	-	-	-	2650	-	-	-	-
2700	-	-	-	-	2700	-	-	-	-
Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	-	-	-	-	1500	-	-	-	-
1550	-	-	-	-	1550	-	-	-	-
1600	-	-	-	-	1600	-	-	-	-
1650	-	-	-	-	1650	-	-	-	-
1700	-	-	-	-	1700	-	-	-	-
1750	-	-	-	-	1750	-	-	-	-
1800	-	-	-	-	1800	-	-	-	-
1850	-	-	-	-	1850	-	-	-	-
1900	-	-	-	-	1900	-	-	-	-
1950	37-26-37 (50)	41-42-17 (62)	-	-	1950	36-34-30 (53)	38-41-21 (58)	-	-
2000	43-20-37 (53)	31-43-26 (52)	-	-	2000	31-28-41 (45)	27-52-21 (53)	-	-
2050	38-27-35 (52)	40-38-22 (59)	-	-	2050	41-24-35 (53)	36-37-27 (54)	-	-
2100	42-25-33 (54)	32-48-20 (56)	-	-	2100	43-23-34 (54)	36-39-25 (56)	-	-
2150	36-23-41 (48)	31-47-22 (54)	-	-	2150	42-23-35 (54)	27-45-28 (50)	-	-
2200	46-13-41 (52)	39-39-22 (58)	-	-	2200	42-21-37 (52)	23-51-26 (48)	-	-
2250	-	-	-	-	2250	-	-	-	-
2300	-	-	-	-	2300	-	-	-	-
2350	-	-	-	-	2350	-	-	-	-
2400	-	-	-	-	2400	-	-	-	-
2450	-	-	40-39-21 (60)	39-38-23 (58)	2450	-	-	25-46-29 (48)	43-30-27 (58)
2500	-	-	38-38-24 (57)	41-40-19 (61)	2500	-	-	40-36-24 (58)	34-44-22 (56)
2550	-	-	44-20-36 (54)	42-29-29 (56)	2550	-	-	40-31-29 (56)	38-38-24 (57)
2600	-	-	46-17-37 (54)	37-23-40 (48)	2600	-	-	46-25-29 (58)	33-37-30 (52)
2650	-	-	41-18-41 (50)	44-22-34 (55)	2650	-	-	47-17-36 (56)	30-35-35 (48)
2700	-	-	45-16-39 (53)	42-22-36 (53)	2700	-	-	38-25-37 (50)	45-21-34 (56)

Figure 8: Full Base-model comparison against Stockfish under anchored planning, grouped by SO-LIS depth and search width. Each cell reports W-D-L and total match score out of 100, highlighting the empirical margins used for the anchored Elo estimates.

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Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	36–32–32 (52)	39–30–31 (54)	–	–	1350	–	–	–	–
1400	23–49–28 (48)	30–32–38 (46)	–	–	1400	–	–	–	–
1450	33–37–30 (52)	24–47–29 (48)	–	–	1450	–	–	–	–
1500	34–38–28 (53)	25–40–35 (45)	–	–	1500	29–40–31 (49)	35–33–32 (52)	–	–
1550	28–38–34 (47)	40–39–21 (60)	–	–	1550	34–35–31 (52)	37–29–34 (52)	–	–
1600	23–47–30 (46)	28–42–30 (49)	–	–	1600	32–36–32 (50)	32–39–29 (52)	–	–
1650	35–29–36 (50)	38–37–25 (56)	–	–	1650	34–30–36 (49)	37–33–30 (54)	–	–
1700	–	–	–	–	1700	36–30–34 (51)	34–35–31 (52)	–	–
1750	–	–	–	–	1750	34–25–41 (46)	34–30–36 (49)	–	–
1800	–	–	35–37–28 (54)	40–28–32 (54)	1800	32–29–39 (46)	32–31–37 (48)	–	–
1850	–	–	32–32–36 (48)	44–29–27 (58)	1850	–	–	–	–
1900	–	–	37–30–33 (52)	45–21–34 (56)	1900	–	–	–	–
1950	–	–	43–30–27 (58)	36–29–35 (50)	1950	–	–	–	–
2000	–	–	35–31–34 (50)	43–27–30 (56)	2000	–	–	46–32–22 (62)	38–27–35 (52)
2050	–	–	37–29–34 (52)	45–27–28 (58)	2050	–	–	37–27–36 (50)	39–26–35 (52)
2100	–	–	36–27–37 (50)	42–34–24 (59)	2100	–	–	29–34–37 (46)	48–17–35 (56)
2150	–	–	–	–	2150	–	–	37–33–30 (54)	39–25–36 (52)
2200	–	–	–	–	2200	–	–	37–31–32 (52)	41–23–36 (52)
2250	–	–	–	–	2250	–	–	36–23–41 (48)	44–20–36 (54)
2300	–	–	–	–	2300	–	–	30–43–27 (52)	38–26–36 (51)
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2500	–	–	–	–	2500	–	–	–	–
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	–	–	–	–	1350	–	–	–	–
1400	–	–	–	–	1400	–	–	–	–
1450	–	–	–	–	1450	–	–	–	–
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	26–45–29 (48)	31–33–36 (48)	–	–	1600	–	–	–	–
1650	37–39–24 (56)	28–32–40 (44)	–	–	1650	–	–	–	–
1700	23–39–38 (42)	29–41–30 (50)	–	–	1700	37–36–27 (55)	34–29–37 (48)	–	–
1750	36–29–35 (50)	38–30–32 (53)	–	–	1750	42–23–35 (54)	37–33–30 (54)	–	–
1800	35–24–41 (47)	33–36–31 (51)	–	–	1800	36–28–36 (50)	31–35–34 (48)	–	–
1850	41–21–38 (52)	30–28–42 (44)	–	–	1850	38–21–41 (48)	33–32–35 (49)	–	–
1900	–	–	–	–	1900	37–27–36 (50)	37–37–26 (56)	–	–
1950	–	–	–	–	1950	43–18–39 (52)	38–24–38 (50)	–	–
2000	–	–	–	–	2000	–	–	–	–
2050	–	–	–	–	2050	–	–	–	–
2100	–	–	–	–	2100	–	–	–	–
2150	–	–	39–23–38 (50)	32–30–38 (47)	2150	–	–	–	–
2200	–	–	33–40–27 (53)	38–25–37 (50)	2200	–	–	30–40–30 (50)	32–34–34 (49)
2250	–	–	36–31–33 (52)	30–27–43 (44)	2250	–	–	29–43–28 (50)	49–26–25 (62)
2300	–	–	32–33–35 (48)	37–28–35 (51)	2300	–	–	31–30–39 (46)	30–35–35 (48)
2350	–	–	35–34–31 (52)	40–25–35 (52)	2350	–	–	26–32–42 (42)	40–34–26 (57)
2400	–	–	28–26–46 (41)	26–38–36 (45)	2400	–	–	39–32–29 (55)	31–32–37 (47)
2500	–	–	–	–	2500	–	–	30–29–41 (44)	28–41–31 (48)

Figure 9: Full Mini-model comparison against Stockfish under anchored planning, grouped by SO-LIS depth and search width. Each cell reports W–D–L and total match score out of 100, showing the anchored scoring margins used for rating estimation.